The Impact of Home Allergen Reduction Kits on Health Outcomes in Individuals with Autoimmune Skin Disorders

Methods for Public Policies Evaluation

Hoxha Enio

Motivation

- Atopic dermatitis (AD) affects an estimated up to 8% of adults in Italy. (Pocetta et al., 2014)
- Environmental allergens and irritants are associated with increased flare frequency and symptom severity. (Silverberg, J.I. 2017)
- Household preventive tools (e.g., anti-mite mattress covers) can be costly, and adoption may be lower among low-income households.
- AD flare-ups increases: healthcare use (Drucker et al. 2017), sleep disruption in children (Chang et al. 2016) and absenteeism (Chung, Simpson 2019).

Research Question

Does providing households with an Allergen-Reduction Kit reduce flare-ups among individuals with atopic dermatitis?

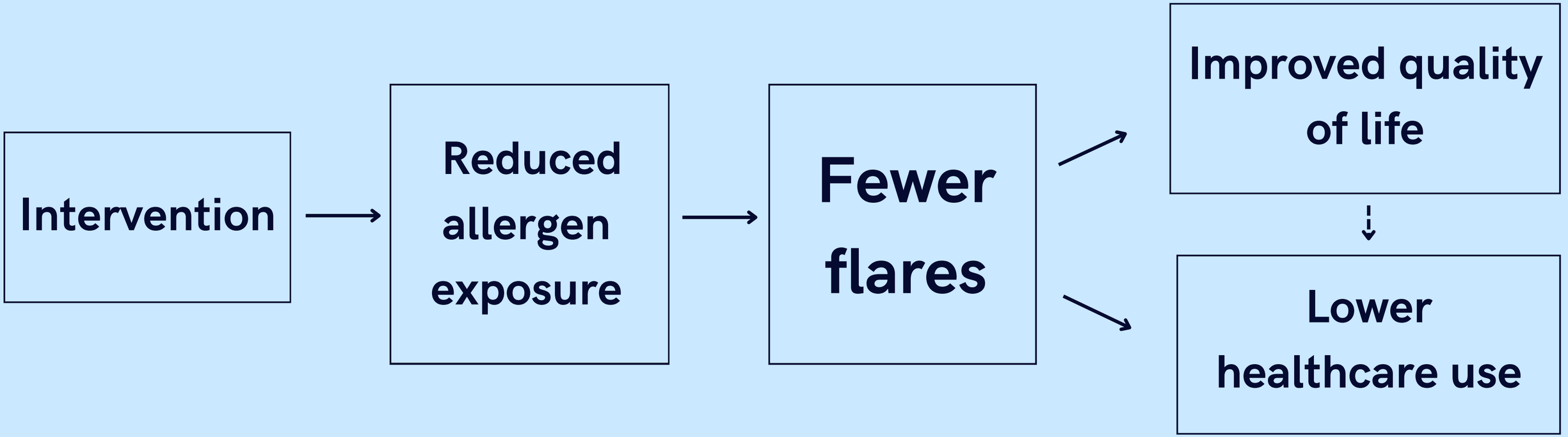
Intervention

Provide a standardised “Allergen Reduction Kit” designed to reduce indoor exposure to common irritants associated with AD flare-ups

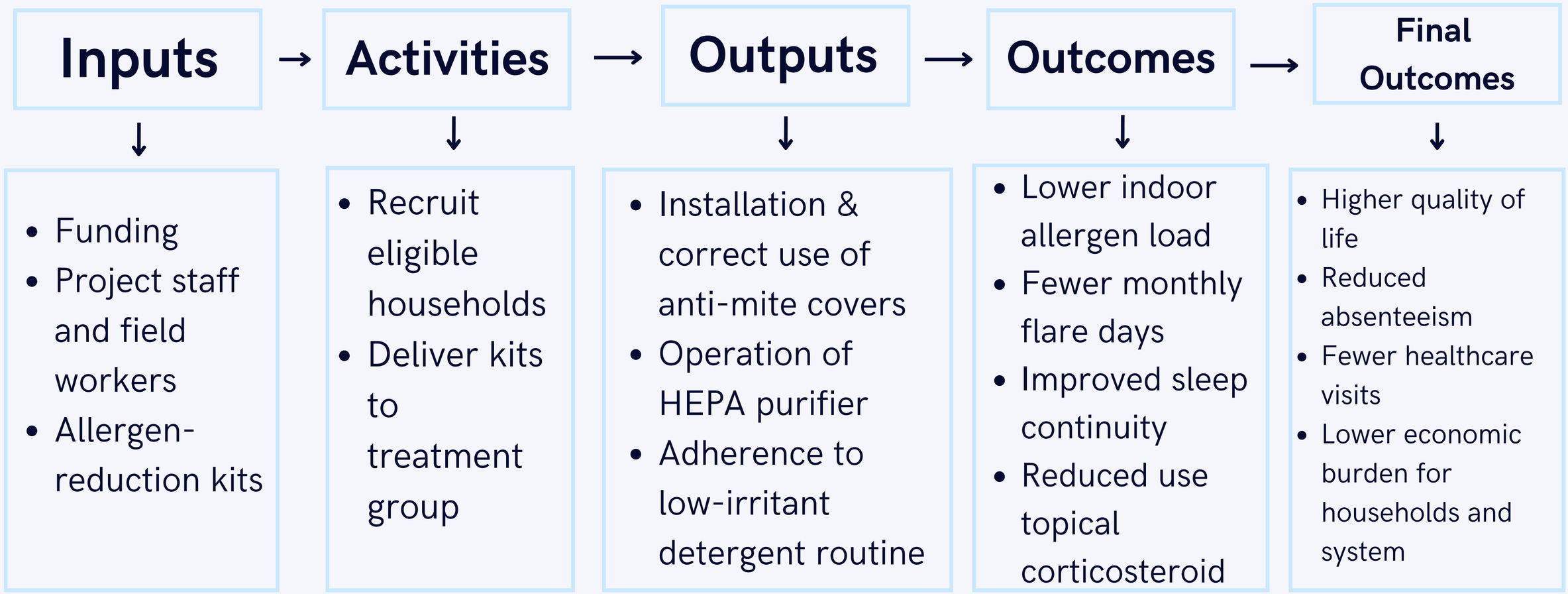
- Anti-mite mattress and pillow covers
- Air Purifier with HEPA filter
- Skin detergent for sensitive skin
- Clear instructions for correct use and environmental allergen reduction routine.



Theory of change



Result chain



Key Performance Indicators

Clinical Outcomes

- Flare days per months
- EASI score (eczema severity)

Economic & System Level Indicators

- Healthcare utilisation (doctor visits, corticosteroid use)
- Out-of-pocket expenses
- Work/School absenteeism

Behavioural & Environmental Indicators

- Use of anti-mite encasings and HEPA purifier
- Adherence to low-irritant detergent routine

Experimental Design (I)

Desing

- Individually randomized controlled trial (1:1)

Unit of Randomization

- Household level (one eligible AD adult per household)

Sample

- Adults diagnosed with atopic dermatitis
- Recruited through dermatology clinics / community centers

Stratification (pre-randomization)

- Baseline severity
- Urban vs rural residence
- Existing use of allergen-control covers

Econometric Strategy

Intent to Treat (ITT)

$$Yi = \alpha + \beta Ti + \sum_{k} \gamma k Xik + \varepsilon i$$

- Yi: outcome (monthly flare count, EASI)
- Ti : treatment assignment (1 = kit offered, 0 = control)
- Xi: pre-specified baseline covariates
- β: Intent-to-Treat (ITT) effect Estimated with OLS robust standard errors

Why ITT?

- Policy relevant: estimates effect of offering the intervention, not usage.

Econometric Strategy (II)

When compliance is imperfect: CACE (LATE)

Estimate the effect among compliers using assignment as an instrument.

First stage: $D_i = \pi_0 + \pi_1 Z_i + \nu_i$

- Z_i: random assignment
- D_i: actual kit usage
- π_1: effect of assignment on uptake

Second stage (2SLS): $Y_i = \alpha + \beta D_i + \varepsilon_i$

- β = CACE (LATE): effect among compliers
- Identification: assignment is exogenous by design

Interpretation

- ITT = effect of being offered the kit
- CACE = effect of actually using the kit

Threats to Internal Validity

Non-Compliance

- Treated households may under use the kit, controls may adopt similar tools.
- Effect: dilutes treatment effect.
- Analytical mitigation: ITT remains unbiased; CACE/LATE estimated via IV.

Observer Bias

- Clinical evaluators blinded to treatment assignment.

Reporting Bias

- Self-reported flare diaries may include measurement noise.
- Mitigated by standardized clinical scales

Other Threats Minimised by Design

- No intra-household spillovers (one participant per household).
- Random assignment ensures balance in observed/unobserved confounders.

External Validity

Contextual Heterogeneity

- Environmental conditions (humidity, dust-mite load)
- Housing quality / ventilation
- Socio-economic status (baseline exposure, ability to maintain routines)

Limitations to Generalizability

- Severe AD patients under systemic therapy
- Regions with very low dust-mite prevalence (dry climates, high altitude)
- Households already using high-quality allergen control
- Cultural differences in cleaning routines and sleep habits

Power

Primary outcome: Monthly flare count.

Target effect: -1 flare/month

Final sample size will be determined after baseline data collection, using the observed mean and variance of monthly flare frequency.

Secondary outcomes:

- Any flare this month (binary): 138 for 15% MDE
- EASI score (continuous).

Scenario	Mean flares/month	SD	Required n/arm
A — Low variability	2	1.5	36
B — Moderate variability	3	2.0	63
C — High variability	4	3.0	141

Time Line and Budget

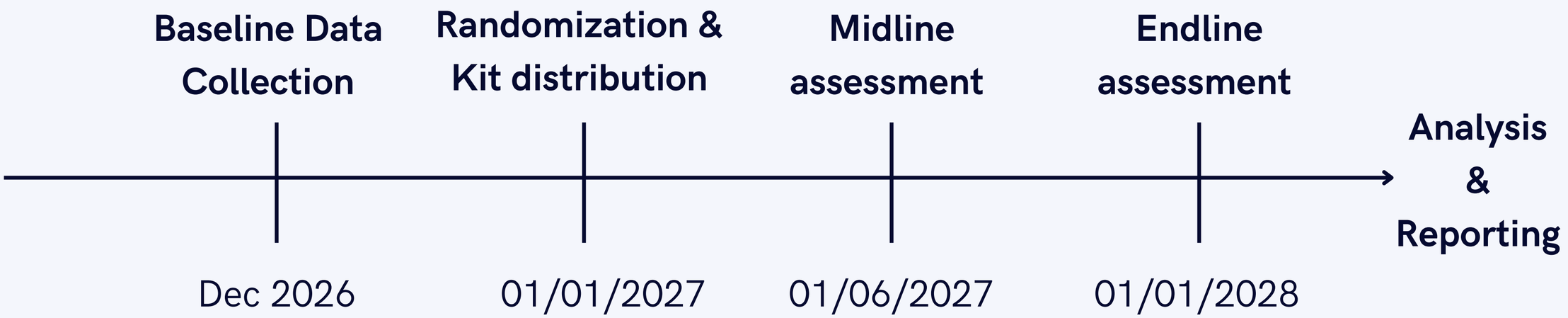
Budget:

Full kit (cover + HEPA purifier + cleanser + leaflet): €260 / kit

Total cost: 282 × € 260 = € 73.320

Staff and Logistics estimate = € 84.360

Total = €157.680



References

Pocetta, G. et al. (2014). Adult eczema in Italy: prevalence and associations with environmental factors. J Eur Acad Dermatol Venereol.

Huang, H.-J., Sarzsinszky, E., Vrtala, S. (2023). House dust mite allergy: importance for diagnosis and immunotherapy. Int Arch Allergy Immunol.

Silverberg, J.I. (2017). Environmental factors and atopic dermatitis. JAMA Dermatology.

Drucker, A.M. et al. (2017). The burden of atopic dermatitis: outpatient visits, medications and costs. J Invest Dermatol.

Chung, J. & Simpson, E.L. (2019). Socioeconomic burden of atopic dermatitis: absenteeism and productivity losses. Ann Allergy Asthma Immunol.

Chang, Y.S. et al. (2016). Sleep disturbance mechanisms in atopic dermatitis. Pediatrics.

Merola, J.F. et al. (2025). Defining flares in atopic dermatitis: narrative review. Am J Clin Dermatol.

Nielsen, M. et al. (2025). Predictors of flares and disease severity using machine learning. JAMA Dermatology.

Conclusion & Questions